Session 8

1. Write a PL/SQL block that declares two variables: one for a user's name (VARCHAR2) and one for their age (NUMBER). Assign values and print both using DBMS_OUTPUT.PUT_LINE.

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```
USE db_restaurant;

DROP PROCEDURE IF EXISTS show_user;

DELIMITER &&

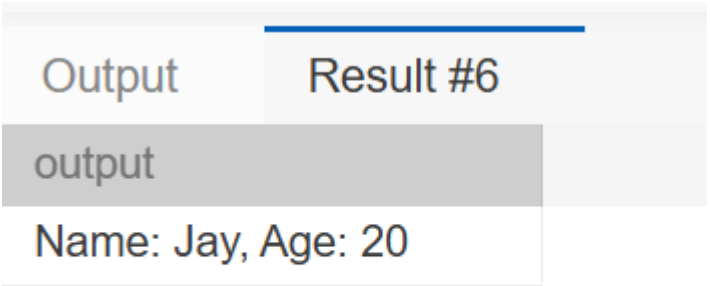
CREATE PROCEDURE show_user()
BEGIN
   DECLARE v_name VARCHAR(255);
   DECLARE v_age INT;

   SET v_name = 'Jay';
   SET v_age = 20;

   SELECT CONCAT('Name: ', v_name, ', Age: ', v_age) AS output;
END &&

DELIMITER ;

CALL show_user();
```



2. Create a PL/SQL block that uses an IF statement to check if a given order amount is above 500. If yes, print 'Eligible for free delivery', else print 'Delivery charges apply.

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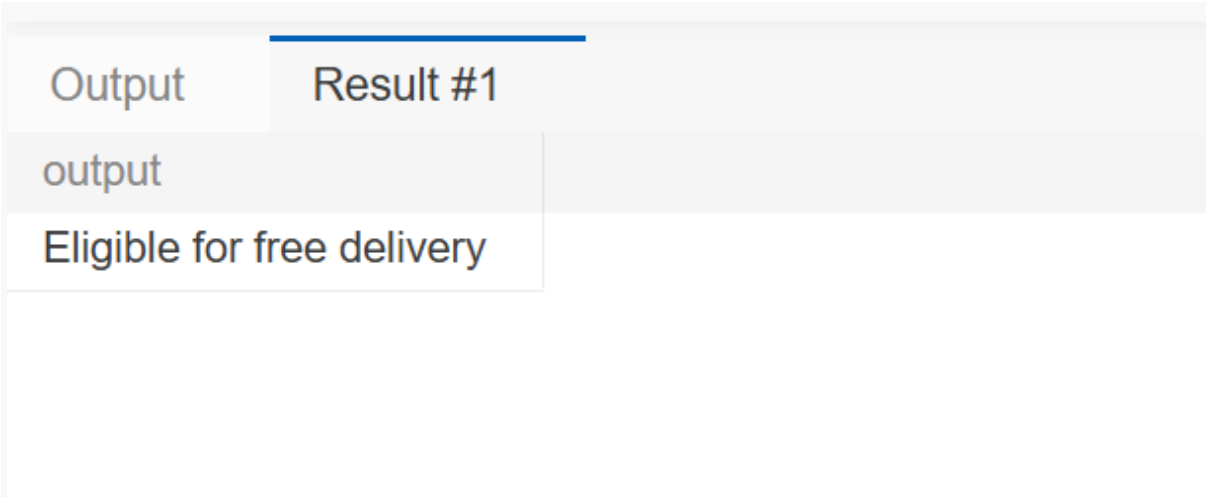
```
USE pl_sql;

DELIMITER &&

CREATE PROCEDURE check_delivery(IN p_order_amount
DECIMAL(10,2))
BEGIN
   IF p_order_amount > 500 THEN
      SELECT 'Eligible for free delivery' AS output;
   ELSE
      SELECT 'Delivery Charges Apply' AS output;
   END IF;
END

DELIMITER ;

CALL check_delivery(1000);
```



3. Write a PL/SQL block that uses a simple LOOP to print the numbers 1 to 5 using DBMS_OUTPUT.PUT_LINE. Hint: Use a counter variable and EXIT WHEN condition inside the loop.

➔

```
USE pl_sql;

DELIMITER &&

CREATE PROCEDURE print_numbers()
BEGIN

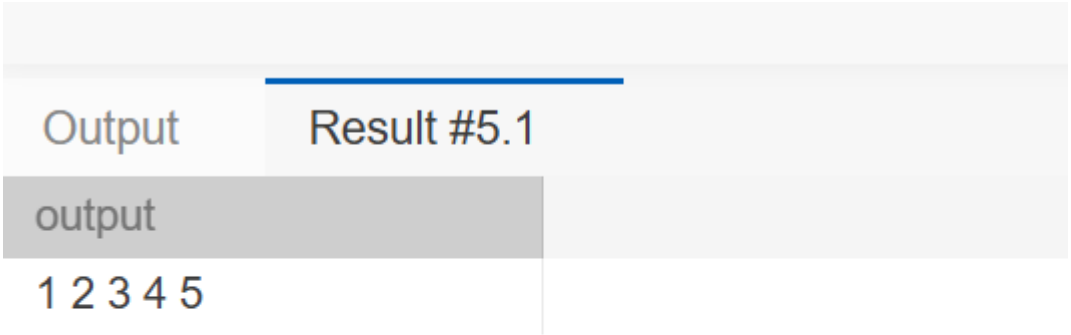
   DECLARE v_counter INT DEFAULT 1;
   DECLARE v_result VARCHAR(50) DEFAULT '';

   simple_loop: LOOP
      SET v_result = CONCAT(v_result, v_counter, ' ');
      SET v_counter = v_counter + 1;
      IF v_counter > 5 THEN
         LEAVE simple_loop;
      END IF;
   END LOOP;

   SELECT v_result AS output;
END

DELIMITER ;

CALL print_numbers();
```



4. Declare a cursor in PL/SQL to select all product names from a table called PRODUCTS. Fetch each product name and print it using DBMS_OUTPUT.PUT_LINE. Hint: Assume the PRODUCTS table has a column PRODUCT_NAME.

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```
USE pl_sql;

CREATE TABLE PRODUCTS (
   product_id INT PRIMARY KEY,
   product_name VARCHAR(100)
);
INSERT INTO PRODUCTS VALUES
(1, 'Laptop'),
(2, 'Phone'),
(3, 'Mouse');

DELIMITER &&

CREATE PROCEDURE show_products()
BEGIN
   DECLARE v_name VARCHAR(100);
   DECLARE v_done INT DEFAULT 0;
   DECLARE v_list TEXT DEFAULT '';
   DECLARE c CURSOR FOR SELECT product_name FROM PRODUCTS;
   DECLARE CONTINUE HANDLER FOR NOT FOUND SET v_done = 1;

   OPEN c;
   FETCH c INTO v_name;
   WHILE v_done = 0 DO
      SET v_list = CONCAT(v_list, v_name, ' ');
      FETCH c INTO v_name;
   END WHILE;
   CLOSE c;

   SELECT v_list AS output;
END

DELIMITER ;

CALL show_products();
```

Output | Result #8.1

output

Laptop Phone Mouse

5. Write a PL/SQL block that tries to divide two numbers entered as variables. If a division by zero occurs, catch the exception and print 'Cannot divide by zero'. Otherwise, print the result.

➔

```
USE pl_sql;

DELIMITER &&

CREATE PROCEDURE safe_divide(IN p_num1 DECIMAL(10,2), IN p_num2
DECIMAL(10,2))

BEGIN
   IF p_num2 = 0 THEN
      SELECT 'Cannot divide by zero' AS output;
   ELSE
      SELECT CONCAT('Result: ', p_num1 / p_num2) AS output;
   END IF;
END

DELIMITER ;

CALL safe_divide(10, 0);
CALL safe_divide(10, 2);
```

Output | Result #5 | Result #6.1

output

Cannot divide by zero

Output | Result #5 | Result #6.1

output

Result: 5.000000